
Notes for ‘Leibniz notation’

Important Ideas and Useful Facts:

- (i) **Leibniz notation and differentials:** An alternative way of expressing the derivative as a limit is the following:

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}$$

where Δx represents a small change in the value of x and Δy as the corresponding small change in the value of $y = f(x)$. With this notation it is common and very useful to write

$$f'(x) = \frac{dy}{dx} \quad \text{and} \quad f'(x)dx = dy ,$$

called *Leibniz notation*, and think of dy and dx as ‘infinitesimally’ small idealised values called *differentials*. We think of the Greek Δ becoming our ordinary d ‘in the limit’.

- (ii) **Some common derivatives and properties using Leibniz notation:**

(a) $\frac{d}{dx}(k) = 0$ for any constant k , $\frac{d}{dx}(x) = 1$, $\frac{d}{dx}(x^2) = 2x$ $\frac{d}{dx}(x^3) = 3x^2$.

(b) $\frac{d}{dx}(x^n) = nx^{n-1}$ for any exponent n (though possibly with restrictions on x).

(c) $\frac{d}{dx}(ky) = k \frac{dy}{dx}$ (constants come out the front).

(d) $\frac{d}{dx}(u + v) = \frac{du}{dx} + \frac{dv}{dx}$ (the derivative is additive).

(e) $\frac{d}{dx}(\sin x) = \cos x$, $\frac{d}{dx}(\cos x) = -\sin x$, $\frac{d}{dx}(\tan x) = \frac{1}{\cos^2 x} = \sec^2 x$.

(f) $\frac{d}{dx}(e^x) = e^x$, $\frac{d}{dx}(\ln x) = \frac{1}{x}$.

- (iii) **Inverting the derivative using Leibniz notation:** If the function $y = f(x)$ is invertible then the rule for the derivative of the inverse function f^{-1} can be deduced by ‘inverting’ $\frac{dy}{dx}$, as though it were an ordinary fraction, to get

$$\frac{dx}{dy} = \frac{1}{dy/dx} ,$$

and expressing the right-hand side in terms of y . If we revert to using x as an input then we produce a rule for the derivative of the inverse function.

Examples and derivations:

1. Find the derivative $\frac{dy}{dx}$ when $y = 4x^3 - x^2 + 6x - 9$.

Solution: We have

$$\frac{dy}{dx} = 4 \frac{d}{dx}(x^3) - \frac{d}{dx}(x^2) + 6 \frac{d}{dx}(x) - \frac{d}{dx}(9) = 4(3x^2) - 2x + 6 = 12x^2 - 2x + 6 .$$

2. Find the derivative $\frac{dy}{dt}$ when $y = 6e^t - 3\ln t + 4t^2$.

Solution: We have

$$\frac{dy}{dt} = 6\frac{d}{dt}(e^t) - 3\frac{d}{dt}(\ln t) + 4\frac{d}{dt}(t^2) = 6e^t - \frac{3}{t} + 8t .$$

3. Find the derivative $\frac{dy}{dx}$ when $y = 2\sin x - 3\cos x$.

Solution: We have

$$\frac{dy}{dx} = 2\frac{d}{dx}(\sin x) - 3\frac{d}{dx}(\cos x) = 2\cos x - 3(-\sin x) = 2\cos x + 3\sin x .$$

4. Find the derivative $\frac{du}{d\theta}$ when $u = \theta^2 + 7\cos \theta - 5\sin \theta$.

Solution: We have

$$\frac{du}{d\theta} = \frac{d}{d\theta}(\theta^2) + 7\frac{d}{d\theta}(\cos \theta) - 5\frac{d}{d\theta}(\sin \theta) = 2\theta - 7\sin \theta - 5\cos \theta .$$

5. We use Leibniz notation to explain why $\frac{d}{dx}(\ln x) = \frac{1}{x}$.

Proof: Put $y = e^x$, so that $x = \ln y$ and, from earlier work,

$$\frac{dy}{dx} = y' = e^x = y .$$

Hence, reciprocating, we get

$$\frac{d}{dy}(\ln y) = \frac{dx}{dy} = \frac{1}{dy/dx} = \frac{1}{y} .$$

Reverting to using x as an input, we get

$$\frac{d}{dx}(\ln x) = \frac{1}{x} ,$$

as required.

6. We use Leibniz notation to explain why $\frac{d}{dx}(\sqrt{x}) = \frac{1}{2\sqrt{x}}$, for $x \geq 0$.

Proof: Put $y = x^2$, so that $x = \sqrt{y}$ and, from earlier work,

$$\frac{dy}{dx} = y' = 2x = 2\sqrt{y} .$$

Hence, reciprocating, we get

$$\frac{d}{dy}(\sqrt{y}) = \frac{dx}{dy} = \frac{1}{dy/dx} = \frac{1}{2\sqrt{y}} .$$

Reverting to using x as an input, we get

$$\frac{d}{dx}(\sqrt{x}) = \frac{1}{2\sqrt{x}} ,$$

as required.

Note that, using fractional power notation, this becomes

$$\frac{d}{dx}(x^{1/2}) = \frac{1}{2}x^{-1/2} ,$$

which is a special case of the formula $\frac{d}{dx}(x^n) = nx^{n-1}$, when $n = \frac{1}{2}$.

7. We use Leibniz notation to explain why $\frac{d}{dx}(\sqrt[3]{x}) = \frac{1}{3x^{2/3}}$.

Proof: Put $y = x^3$, so that $x = \sqrt[3]{y} = y^{1/3}$ and, from earlier work,

$$\frac{dy}{dx} = y' = 3x^2 = 3y^{2/3} .$$

Hence, reciprocating, we get

$$\frac{d}{dy}(\sqrt[3]{y}) = \frac{dx}{dy} = \frac{1}{dy/dx} = \frac{1}{3y^{2/3}} .$$

Reverting to using x as an input, we get

$$\frac{d}{dx}(\sqrt[3]{x}) = \frac{1}{3x^{2/3}} ,$$

as required.

Note that, using fractional power notation, this becomes

$$\frac{d}{dx}(x^{1/3}) = \frac{1}{3}x^{-2/3} ,$$

which is a special case of the formula $\frac{d}{dx}(x^n) = nx^{n-1}$, when $n = \frac{1}{3}$.